

PREDICTIVE QUALITY CONTROL REPORT

Author: Ubaid ur Rahman: Analytical Chemist & Data Analyst
Target Audience: Plant Operations, QC Directors, & Executive Leadership
Date: June 2026

Waiting for traditional laboratory testing creates severe production bottlenecks. If a batch drops below quality standards during processing, it is often noticed only after expensive bottling and packaging phases, leading to significant material scrap and lost revenue. This project demonstrates an automated predictive engine that catches defects early.

Executive Summary: By shifting from reactive manual sampling to an automated, data-driven machine learning inspector, the facility can halt out-of-specification batches during fermentation, saving an estimated **\$11,500 per deflected incident**.

1. Statistical Verification (Proof of Concept)

An Ordinary Least Squares (OLS) regression analysis was executed to prove that specific chemical properties dictate final product quality. The analysis confirms that Volatile Acidity and Alcohol content are primary quality drivers, eliminating the risk of random data correlation.

0.3242	< 0.001	1,358
R-SQUARED VALUE	P-VALUE SIGNIFICANCE	VALID OBSERVATIONS

Interpretation for Executive Leadership

- **R-Squared (32.4%):** These two key chemical components control over 32% of the total variance in final product quality. For a complex biological process, this represents an exceptionally high level of predictability.
- **P-Value Significance:** The probability of these findings happening by chance is essentially zero. This gives management mathematical certainty for operational changes.

2. Automated Machine Learning Pipeline

The code is engineered to run completely offline within a local environment. It functions as an automated inspector through the following four systematic phases:

Pipeline Stage	Technical Action Taken	Operational Benefit
1. ETL & Cleaning	Automatically reads raw data, strips out duplicate sensor readings, and removes incomplete logs.	Guarantees baseline data hygiene without manual spreadsheet cleanup.
2. Chemometrics	Calculates a custom preservative stability metric: Free sulfur dioxide divided by Total sulfur dioxide.	Isolates active chemical compounds before running predictions.
3. AI Classification	Trains an ensemble machine learning algorithm to classify batches into High Quality or Low Quality.	Replaces slow human taste panels with instant, objective digital scoring.
4. SQL Database	Streams all chemical features along with the AI predictions straight into a central SQL database.	Provides live data infrastructure for downstream managerial tracking dashboards.

3. Financial Return on Investment (ROI)

The financial framework evaluates the cost-benefit of catching an out-of-specification batch during early fermentation versus catching it reactively after packaging.

Production Stage	Operational Activity	Expense Without AI	Expense With AI	Net Savings
Fermentation	Early intervention and recipe adjustment	\$0	\$1,500	-\$1,500
Packaging	Scrapping ruined bottles, labels, and boxes	\$13,000	\$0	\$13,000
Total Per Incident	Operational Impact Analysis	\$13,000	\$1,500	\$11,500 Saved